

Erdős Problem #1101

EP-1101

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

If $u = \{u_1 < u_2 < \dots\}$ is a sequence of integers such that $(u_i, u_j) = 1$ for all $i \neq j$ and $\sum \frac{1}{u_i} < \infty$ then let $\{a_1 < a_2 < \dots\}$ be the sequence of integers which are not divisible by any of the u_i . For any x define t_x by $u_1 \cdots u_{t_x} \leq x < u_1 \cdots u_{t_x} u_{t_x+1}$. We call such a sequence u_i good if, for all $\epsilon > 0$, if x is sufficiently large then $\max_{a_k < x} (a_{k+1} - a_k) < (1 + \epsilon) t_x \prod_i \left(1 - \frac{1}{u_i}\right)^{-1}$. Is there a good sequence such that $u_n < n^{O(1)}$? Is there a good sequence such that $u_n \leq e^{o(n)}$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>